

Chapter 9: Statics and Torque

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PHY201
Lecture 9



Part I

Static Equilibrium

Outline of Part I: Static Equilibrium

Introduction to Statics

First Condition for Equilibrium

Torque

Second Condition for Equilibrium

What is Statics?

Equilibrium: $\vec{F}_{\text{net}} = m\vec{a} = 0$ (linear) $\vec{\tau}_{\text{net}} = I\vec{\alpha} = 0$ (rotational)

An object is in **equilibrium** whenever the net force *and* the net torque are simultaneously zero ($\vec{a} = 0$).

Static Equilibrium

- ▶ $\vec{v} = 0$ (at rest)
- ▶ $\vec{a} = 0$ (remains at rest)

Dynamic Equilibrium

- ▶ $\vec{v} \neq 0$ (in motion)
- ▶ $\vec{a} = 0$ (remains in motion)

Statics focuses on static equilibrium. Every structure you rely on — buildings, bridges, bones — must not accelerate *linearly or rotationally*.

Two Conditions for Static Equilibrium

First Condition

No translational acceleration:

$$\vec{F}_{\text{net}} = 0 \quad \Rightarrow \quad \begin{cases} \sum F_x = 0 \\ \sum F_y = 0 \end{cases}$$

Second Condition

No rotational acceleration:

$$\vec{\tau}_{\text{net}} = 0 \quad \Rightarrow \quad \sum \tau = 0$$

Both conditions must hold *simultaneously* for static equilibrium.

An object can satisfy the first condition yet still rotate (e.g., a couple of equal and opposite forces not on the same line of action).

Outline of Part I: Static Equilibrium

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First Condition for Equilibrium

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Second Condition for Equilibrium

First Condition for Equilibrium

First Condition: $\sum \vec{F} = 0$

The vector sum of all external forces acting on an object must be zero:

$$\sum F_x = 0 \quad \text{and} \quad \sum F_y = 0$$

- ▶ This is Newton's First Law applied to both directions simultaneously.
- ▶ It prevents the object from *accelerating* in any direction.
- ▶ It does *not* prevent the object from rotating.

Block on Surface

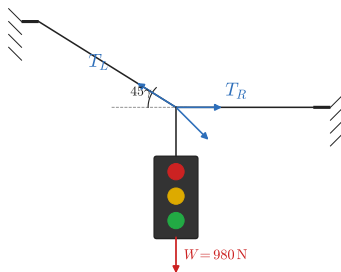
$$\sum F_y = N - W = 0$$

~~*Two-Rope Hang*~~

$$\sum F_x = 0, \sum F_y = 0$$

Example: First Condition (cp2e_ch9_ex1)

A traffic light of weight $W = 980 \text{ N}$ is supported by two cables. The left cable makes 45.0° with the horizontal; the right cable is horizontal. Find the tension in each cable.



$$\sum F_y = 0 : T_L \sin 45^\circ - W = 0$$

The angled cable supports all the

Check Your Understanding

A picture frame of weight 50.0 N hangs from two wires that each make an angle of 30.0° with the horizontal.

What is the tension in each wire?

Come to lecture to see the answer.

Outline of Part I: Static Equilibrium

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Second Condition for Equilibrium

Torque: The Rotational Analogue of Force

Just as **force** causes linear acceleration, **torque** causes rotational (angular) acceleration.

Torque is the tendency of a force to rotate an object about a pivot point.

- ▶ Pushing a door near the hinges requires more force than pushing near the edge.
- ▶ A longer wrench makes loosening a bolt easier.
- ▶ These are everyday examples of torque — the same force applied farther from the pivot produces more torque.

torquēre ['tor.k^we:re]

verb – Latin

to twist, to wrench

Descendants: **torque**, “torture”, “contort”, “distort”

Torque: Definition and Formula

Torque (τ) — SI Unit: newton·meter ($\text{N} \cdot \text{m}$)

The torque produced by a force $\vec{F}$ applied at position $\vec{r}$ from the pivot is:

$$\tau = rF \sin \theta_{\vec{r}\vec{F}}$$

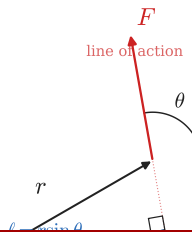
where r is the distance from the pivot to where the force is applied, F is the magnitude of the force, and $\theta_{\vec{r}\vec{F}}$ is the angle between $\vec{r}$ and $\vec{F}$.

Lever Arm ℓ (Moment Arm)

Factoring $\tau = rF \sin \theta_{\vec{r}\vec{F}}$:

$$\tau = \underbrace{(r \sin \theta_{\vec{r}\vec{F}})}_{\ell} F = \ell F$$

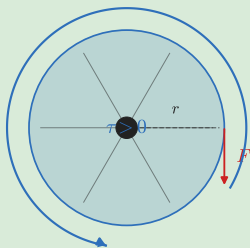
ℓ is the **lever arm**: drawn from the pivot point to where the force is applied



Sign Convention for Torque

Counterclockwise (CCW) = Positive

τ



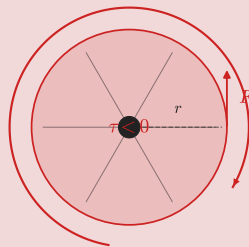
Counterclockwise (CCW) = $+\tau$

Force

tends to rotate the object
counterclockwise

$$\Rightarrow \tau > 0$$

Clockwise (CW) = Negative τ



Clockwise (CW) = $-\tau$

Force

tends to rotate the object **clockwise**

$$\Rightarrow \tau < 0$$

Example: Calculating Torque

A force of 25.0 N is applied to a wrench at a distance of 0.300 m from the bolt.

- (a) Find the torque if the force is applied perpendicular to the wrench.
- (b) Find the torque if the force is applied at 60.0° to the wrench.

$$\begin{aligned}\text{(a)} \quad \tau &= rF \sin 90^\circ \\ &= (0.300 \text{ m})(25.0 \text{ N})(1)\end{aligned}$$

$$\tau = 7.50 \text{ N} \cdot \text{m}$$

$$\begin{aligned}\text{(b)} \quad \tau &= rF \sin 60^\circ \\ &= (0.300 \text{ m})(25.0 \text{ N})(0.866)\end{aligned}$$

$$\tau = 6.50 \text{ N} \cdot \text{m}$$

Applying the force perpendicular to the wrench maximizes torque. Angling the force reduces its effectiveness.

Check Your Understanding

A force of 40.0 N is applied at the end of a 0.500 m rod, perpendicular to the rod.

(a) What is the torque about the pivot at the other end of the rod?

(b) If the force is instead applied parallel to the rod, what is the torque?

Come to lecture to see the answer.

Check Your Understanding

You are trying to loosen a stuck bolt with a wrench.

Describe two different ways you could increase the torque you apply to the bolt, without increasing the magnitude of the force.

Come to lecture to see the answer.

Outline of Part I: Static Equilibrium

Introduction to Statics

First Condition for Equilibrium

Torque

Second Condition for Equilibrium

Second Condition for Equilibrium

Second Condition: $\sum \tau = 0$

For an object in rotational equilibrium, the net torque about *any* pivot point must be zero:

$$\sum \tau = 0$$

Equivalently: the sum of all CCW torques equals the sum of all CW torques.

- ▶ This ensures the object has no angular acceleration ($\alpha = 0$).
- ▶ This condition must hold about **any** chosen pivot point (not just a special one).
- ▶ Choosing the pivot at an unknown force's point of application eliminates that force from the torque equation.

Choosing the Pivot Point Strategically

Key Insight

A force that acts at the pivot point produces **zero torque** about that pivot ($r = 0$).

∴ Place the pivot at the point where an *unknown* force acts $\Rightarrow$ eliminate it from the torque equation.

Strategy

- ▶ If you have two unknown forces, place the pivot at one of them.
- ▶ The torque equation will then contain only *one* unknown.
- ▶ Solve the torque equation, then use $\sum F = 0$ to find the second unknown.

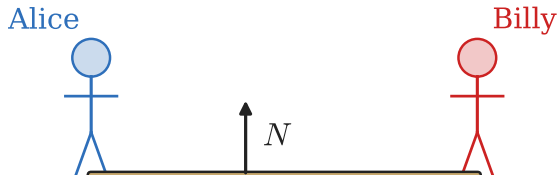
The physics does not change with your choice of pivot — equilibrium holds about any point. You are free to choose whichever makes the algebra simplest.

Example: Seesaw — Find the Unknown Mass

Alice ($m_A = 40.0 \text{ kg}$) and Billy sit on opposite sides of a seesaw. Alice sits $r_A = 1.00 \text{ m}$ from the fulcrum; Billy sits $r_B = 1.50 \text{ m}$ from the fulcrum. Find Billy's mass m_B so the seesaw is in static equilibrium.

$$m_A = 40.0 \text{ kg}$$

$$m_B = 26.7 \text{ kg}$$



Pivot Choice: Same Answer

Equilibrium holds about *any* pivot. Verify the seesaw result by choosing a *different* pivot point.

2. Pivot at Alice's seat (left end):

- ▶ $\tau_A = 0$ ($r_A = 0$ at the pivot)
- ▶ N acts 1.00 m right $\Rightarrow$ CCW (+)
- ▶ W_B acts 2.50 m right $\Rightarrow$ CW (-)

$$\sum \tau = 0 : \quad N(1.00) - m_B g(2.50) = 0$$

$$\text{Sub } N = (m_A + m_B)g :$$

$$(m_A + m_B)(1.00) = m_B(2.50)$$

$$m_A = 1.50 m_B$$

40.0

Same answer!

Pivot choice is a *mathematical* convenience, not a physical constraint.

Best strategy: place the pivot where an *unknown* force acts so it contributes zero torque and drops out of the equation.

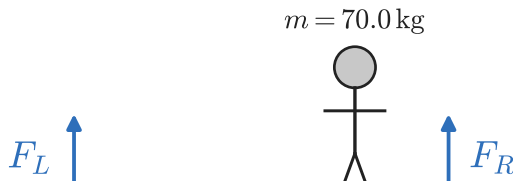
Check Your Understanding

Two children sit on a seesaw pivoted at its center. Child A (35.0 kg) sits 1.20 m from the center. How far from the center must Child B (42.0 kg) sit to balance the seesaw?

Come to lecture to see the answer.

Example: Person on a Plank (cp2e_ch9_ex2)

A uniform plank ($m_p = 30.0$ kg, $L = 4.00$ m) is supported at both ends (left support F_L , right support F_R). A person ($m = 70.0$ kg) stands 1.00 m from the right end. Find F_L and F_R .



Check Your Understanding

A uniform 4.00 m seesaw has mass 20.0 kg. Child A (30.0 kg) sits at the left end; Child B (40.0 kg) sits at the right end. The seesaw pivots at its center.

Is the seesaw in rotational equilibrium? If not, which way does it rotate?

Come to lecture to see the answer.

Check Your Understanding

Where must Child B (40.0 kg) sit from the center of the seesaw to balance Child A (30.0 kg) sitting 2.00 m from the center?

Come to lecture to see the answer.

Check Your Understanding

Can an object satisfy the first condition for equilibrium ($\sum \vec{F} = 0$) but *not* the second ($\sum \tau = 0$)?

Give an example.

Come to lecture to see the answer.

Part II

Applications of Statics

Outline of Part II: Applications of Statics

Problem-Solving Strategy

Stability

Simple Machines

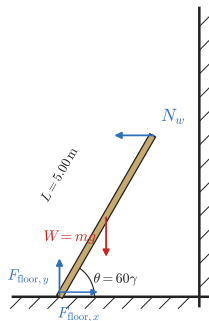
Forces and Torques in Muscles and Joints

Problem-Solving Strategy for Statics

1. **Draw a free-body diagram.** Show every external force acting on the object with arrows indicating direction.
2. **Choose a coordinate system.** The conventional $+x$ right, $+y$ up is usually best.
3. **Identify all forces** and resolve them into x - and y -components.
4. **Choose a pivot point.** Place it where an unknown force acts to simplify the torque equation.
5. **Write the three equilibrium equations:**
$$\sum F_x = 0, \quad \sum F_y = 0, \quad \sum \tau = 0$$
6. **Solve** the system of equations for the unknowns.
7. **Check** your answers for physical reasonableness.

Example: Beam Leaning Against a Wall (cp2e_ch9_ex3)

A uniform beam ($m = 40.0 \text{ kg}$, $L = 5.00 \text{ m}$) leans against a frictionless wall at $\theta = 60.0^\circ$ from the floor. Find the normal force from the wall (N_w) and the reaction at the floor.



Pivot at floor contact (eliminates floor forces):

$$\sum F_x = 0: F_{\text{floor}, x} = N_w = 113 \text{ N}$$

Check Your Understanding

In the beam-against-wall problem, what happens to the force the wall must exert as the angle θ (measured from the floor) increases toward 90° ?

What happens as θ decreases toward 0° ?

Come to lecture to see the answer.

Outline of Part II: Applications of Statics

Problem-Solving Strategy

Stability

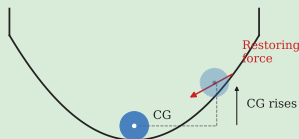
Simple Machines

Forces and Torques in Muscles and Joints

Types of Equilibrium

Stable

Stable Equilibrium

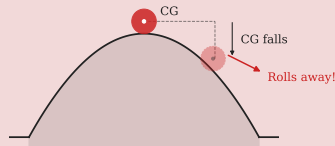


Displaced $\Rightarrow$ returns to equilibrium.

CG *rises* when displaced.

Unstable

Unstable Equilibrium

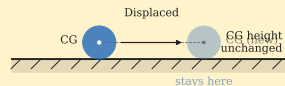


Displaced $\Rightarrow$ moves *away* from equilibrium.

CG *falls* when displaced.

Neutral

Neutral Equilibrium



Displaced $\Rightarrow$ stays in new position.

CG height *unchanged*.

Center of Gravity and Stability

Center of Gravity (CG)

The **center of gravity** is the point at which the weight of an object can be considered to act.

For uniform gravity: CG coincides with the **center of mass**.

Stability Rule

An object is in **stable** equilibrium as long as the vertical line through its CG falls *within* its base of support.

If it falls *outside* the base, the object tips over.

Check Your Understanding

A race car is designed with a very low center of gravity and wide wheelbase. A bus, by contrast, has a high center of gravity and a relatively narrow wheelbase.

Explain why race cars rarely tip over in corners, while loaded buses and trucks occasionally do.

Come to lecture to see the answer.

Check Your Understanding

A person leans forward while carrying a heavy backpack. Why do they lean forward, and why would leaning backward cause them to fall?

Come to lecture to see the answer.

Outline of Part II: Applications of Statics

Problem-Solving Strategy

Stability

Simple Machines

Forces and Torques in Muscles and Joints

Simple Machines and Mechanical Advantage

Mechanical Advantage (MA)

A **simple machine** uses torque balance to allow a small input force to produce a larger output force:

$$MA = \frac{F_{\text{out}}}{F_{\text{in}}} = \frac{d_{\text{in}}}{d_{\text{out}}}$$

where d_{in} and d_{out} are the distances (or lever arms) for the input and output forces.

$MA > 1$: machine amplifies force (at the cost of distance traveled).

$MA < 1$: machine amplifies distance/speed (at the cost of force).

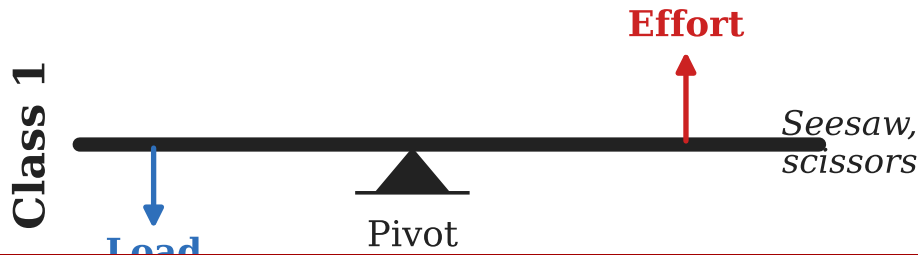
Real machines: $MA_{\text{actual}} < MA_{\text{ideal}}$ due to friction.

Common Simple Machines

► **Lever** (3 classes), **Wheel and Axle**, **Pulley**

Classes of Levers

Class	Pivot location	MA	Examples
1st	Between effort and load	> 1 , $= 1$, or < 1	Seesaw, scissors, pliers
2nd	Load between pivot and effort	Always > 1	Wheelbarrow, nutcracker
3rd	Effort between pivot and load	Always < 1	Tweezers, forearm, broom



Check Your Understanding

A lever has an input arm of 1.20 m and an output arm of 0.30 m.

(a) What is its mechanical advantage?

(b) If the output force needed is 400 N, what input force is required?

Come to lecture to see the answer.

Outline of Part II: Applications of Statics

Problem-Solving Strategy

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Simple Machines

Forces and Torques in Muscles and Joints

Torques in the Human Body

The human body is a system of levers. Bones are the levers, joints are the pivots, and muscles provide the forces.

- ▶ Most muscles attach close to the joint (short lever arm for the muscle).
- ▶ The load (weight of limb + held object) acts farther from the joint.
- ▶ $\therefore$ muscles must exert forces *much larger* than the weight being supported.
- ▶ This design favors **speed and range of motion** over force.

3rd-class lever: effort between pivot and load

Example: Forearm as a Lever (cp2e_ch9_ex4)

A person holds a 10.0 kg ball in their hand. The forearm (mass 2.50 kg) is horizontal. The bicep attaches 4.00 cm from the elbow. The forearm's CG is 16.0 cm from the elbow, and the ball is 38.0 cm from the elbow. Find the bicep force F_B .

Pivot at elbow joint (eliminates joint force):

$$\begin{aligned}\sum \tau_{\text{elbow}} = 0 : & \quad F_B(0.0400) \\ & \quad - (2.50)(9.80)(0.160) \\ & \quad - (10.0)(9.80)(0.380) = 0\end{aligned}$$

$$F_B(0.0400) = 3.92 + 37.24 = 41.16$$

$$F_B = 1029 \text{ N} \approx 1030 \text{ N}$$

The bicep must exert 1030 N — over **ten times** the weight of the ball!

$$MA = \frac{F_{\text{out}}}{F_{\text{in}}} = \frac{(10.0)(9.80)}{1030} \approx 0.095$$

As expected for a 3rd-class lever.

Check Your Understanding

A person stands on their tiptoes. Their calf muscle (Achilles tendon) pulls upward on the heel, and the ball of the foot pushes down on the floor.

Is the foot a first-, second-, or third-class lever in this scenario?

Does the calf muscle exert more or less force than the person's body weight? Explain.

Come to lecture to see the answer.

Check Your Understanding

Why is it harder to hold a heavy grocery bag with your arm fully extended than with your arm bent at the elbow?

Come to lecture to see the answer.

Chapter Summary

Key Equations

- ▶ $\sum F_x = 0$ (1st condition)
- ▶ $\sum F_y = 0$ (1st condition)
- ▶ $\sum \tau = 0$ (2nd condition)
- ▶ $\tau = rF \sin \theta_{\vec{r}\vec{F}}$
- ▶ $MA = \frac{F_{\text{out}}}{F_{\text{in}}} = \frac{d_{\text{in}}}{d_{\text{out}}}$

Key Concepts

- ▶ Both conditions must hold for static equilibrium.
- ▶ Torque depends on force, distance, *and* angle.
- ▶ Choose the pivot to eliminate unknown forces.
- ▶ Stability: CG must be above the base of support.
- ▶ Human body: muscles in 3rd-class lever arrangements exert forces much larger than the load.